

ACADEMY OF TECHNOLOGY

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Power Point Presentation

on

Design and Implementation of an Automated Antenna Radiation
Pattern Measurement System using Embedded Control and
Image Processing

Presented by

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Introduction

- Antenna radiation pattern measurement is important in RF communication systems.
- Traditional methods are manual and time-consuming.
- Manual measurement often causes human error.
- Our project automates rotation, data capture and plotting.
- Uses embedded control and image processing.

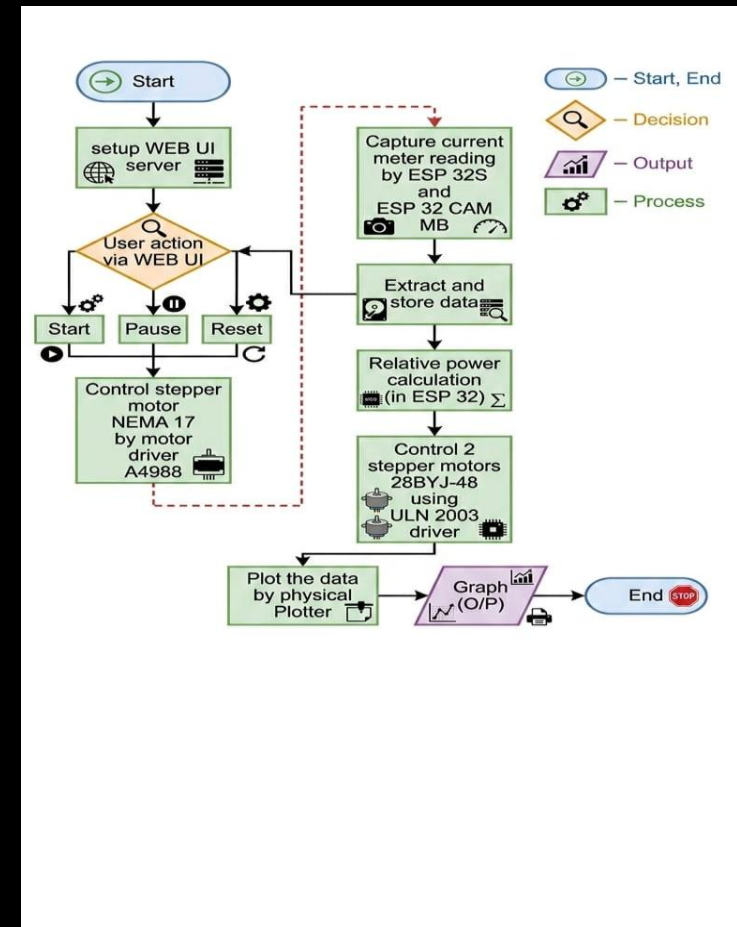
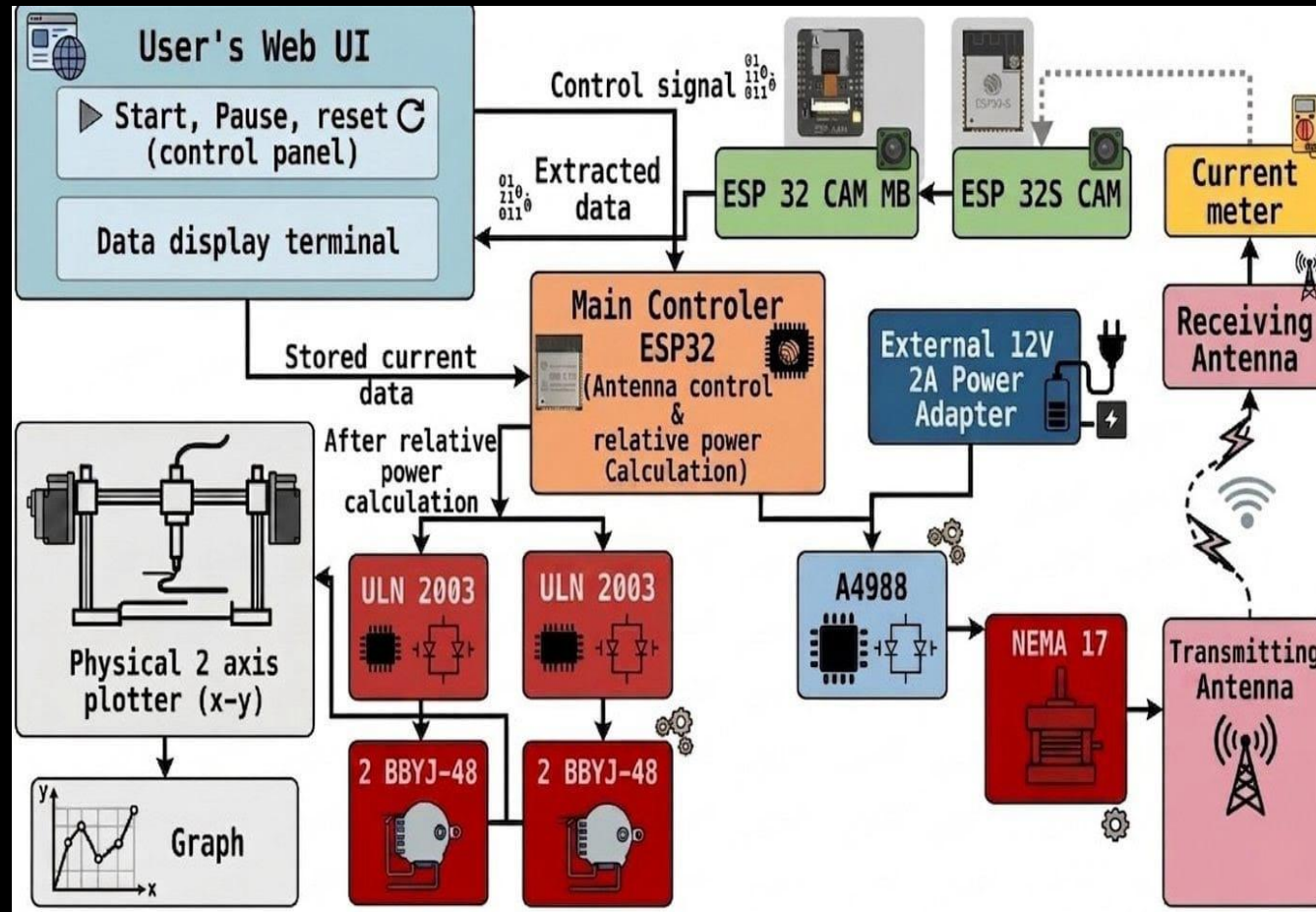
Problem Statement

- Existing systems require manual antenna rotation.
- Meter readings are taken manually.
- Error occur during graph plotting.
- Requires more time and human effort.
- Lack of fully integrated automation.

Objective

- Automate antenna rotation in 10° steps.
- Capture meter readings using ESP32-CAM.
- Extract values using OCR.
- Calculate power in dB.
- Plot radiation pattern automatically on paper.

Block Diagram and Workflow of the System



Technology Used

- Automated and Control Technology.
- Computer Vision and Image Processing.
- Plotting Mechanism

Hardware Components

1. Central Processing Unit
 - ESP 32 (512MB RAM) - Main controller
2. Mechanical Rotation System
 - NEMA 17 Stepper Motor - Rotates antenna
 - A4988 Motor Driver - Controls antenna rotation motor
3. Computer Vision and Image Acquisition
 - ESP 32S CAM - Captures meter images
 - ESP 32 CAM MB - Connector
4. Physical Plotting System
 - 28BYJ-48 Stepper Motor(X-axis) - Horizontal movement
 - ULN2003 Driver Board (X-axis) - X-axis motor control
 - 28BYJ-48 Stepper Motor (Y-axis) - Vertical movement
 - ULN2003 Driver Board (Y-axis) - Y-axis motor control
 - Linear Rails (2 units) - 300mm aluminium guides
 - Timing Belts - GT2 timing belts with pulleys
 - SG90 Micro Servo - Pen lift mechanism
 - Pen Holder - Spring-loaded mechanism
 - Drawing Surface - A4 paper on rigid backing
 - Frame Structure - Acrylic/aluminium frame
5. Power Supply System
 - 12V 4A Power Adapter - USB-C connector for ESP 32

Gap Analysis

- ✓ **Lack of Fully Integrated Automation:** Existing systems automate either rotation OR measurement not both together.
- ✓ **Dependency on Electrical Interface:** Most systems require wired connection to meters, making them less flexible.
- ✓ **Absence of Self-contained Embedded Operation:** Many setups rely on PC-based processing.
- ✓ **No Direct Physical Visualization:** Graphs are digitally plotted but not physically drawn.
- ✓ **Human Intervention Still Required:** In positioning, reading, or plotting stage.

Limitation

- The camera is positioned with proper lighting and focus to capture clear images of the meter display.
- The 30-second stabilization period is sufficient for current reading to become stable at each position.

Feasibility Analysis

- Technical Feasibility
 - NEMA 17 Stepper motors ensure precise angular control.
 - ESP 32S CAM enables non-contact data capture.
 - CNC machine plotter plots the graph physically.
- Operational Feasibility
 - Fully automated operation via single switch.
 - Minimal human supervision required.
 - Does not require complex RF interfacing.
- Economic Feasibility
 - Uses low-cost and easily available components.
 - Avoids expensive antenna measurement instruments.

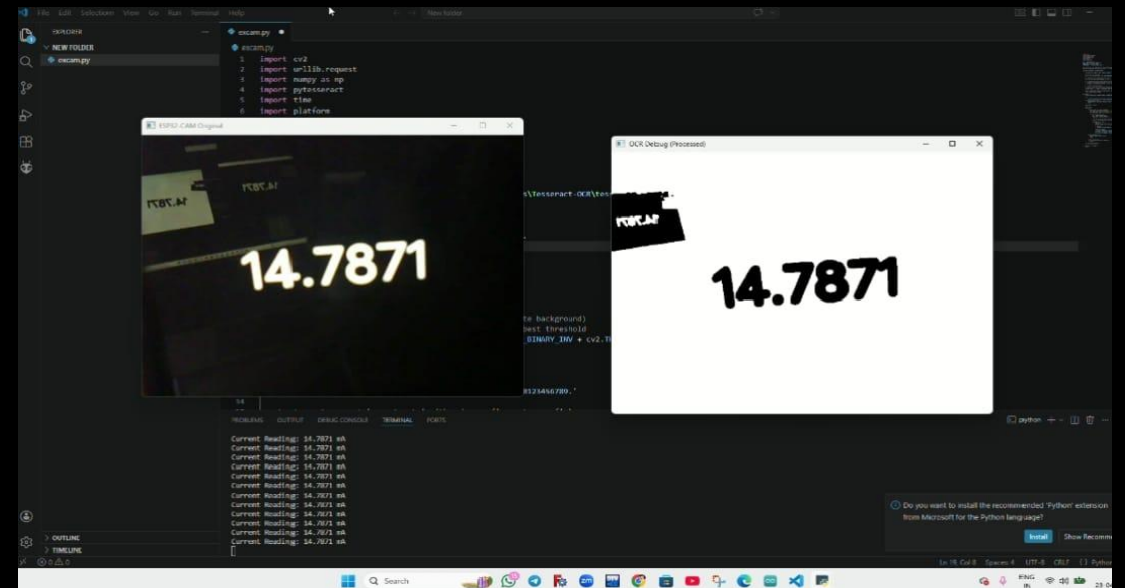
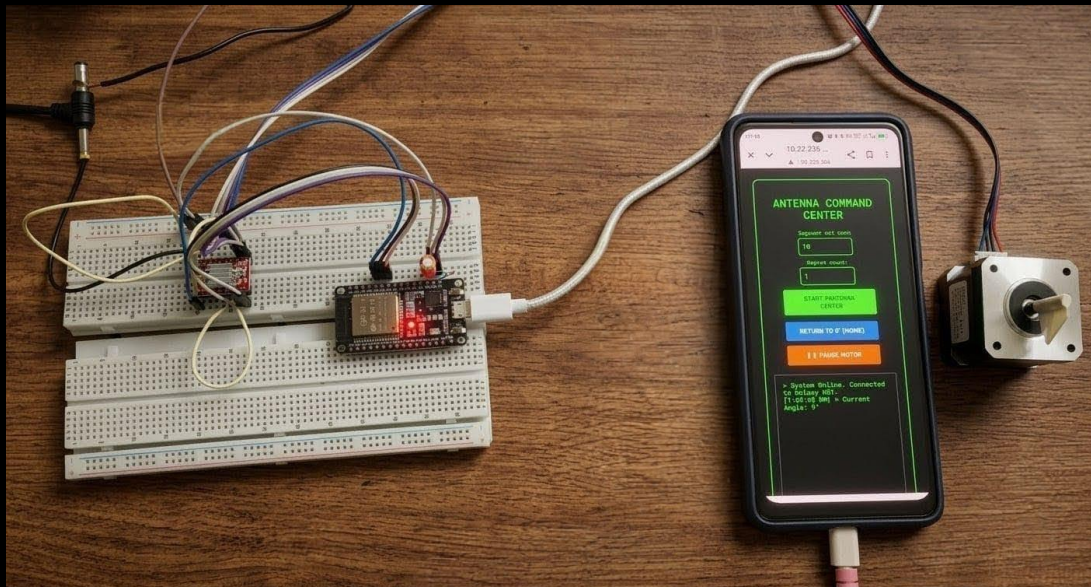
Target Customer

- IoT Wireless Module and Sensor Developers.
- Automated labs.
- Educational Training Kit Manufacturers.

Future Scope

- Polar graph plotting.
- AI-based OCR improvement.

Prototype Development



Conclusion

Till now the antenna rotation using the stepper motor, web-based control, and OCR-based meter reading have been implemented and tested. The complete module will demonstrate the feasibility, accuracy, and practical applicability of the proposed embedded automation system.

Thank You